

ORIGINAL ARTICLE

Propeller flaps in soft tissue sarcoma's defects: Analysis and discussion of current literature

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ABSTRACT

Background and Aim: Propeller flaps have been described as local flaps for soft tissue reconstruction in various anatomical regions, defined as “an island flap that reaches the recipient site through an axial rotation.” The distinctive feature of propeller flaps compared to other pedicled flaps lies in their rotation around a central pivot point. Due to their versatility, ability to achieve like-to-like reconstruction, and minimal donor-site morbidity, propeller flaps have gained widespread use in reconstructive surgery for traumatic defects, tumor excisions, burns, and infections. The aim of this study was to evaluate the outcomes of propeller flaps in oncological reconstruction and to compare them with free flap procedures.

Methods: A retrospective analysis was conducted on 38 patients affected by soft tissue sarcomas who underwent reconstructive surgery following tumor excision. Nineteen patients (50%) were treated with propeller flaps and nineteen (50%) with free flaps. Clinical outcomes, including flap survival, complication rates, and salvage procedures, were assessed and compared between the two groups.

Results: Both propeller and free flaps achieved satisfactory reconstructive outcomes with comparable rates of flap survival. Propeller flaps demonstrated reduced operative time and donor site morbidity, while maintaining acceptable safety and reliability in oncological settings. Complications were effectively managed with conservative or minor surgical interventions, and overall salvage rates were consistent with data reported in the literature.

Conclusions: Propeller flaps represent a reliable and versatile reconstructive option in oncological soft tissue surgery, offering a valid alternative to free flaps in selected patients. Their use allows adequate defect coverage



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with minimal donor site morbidity and satisfactory functional and aesthetic outcomes, supporting their role as a safe and efficient technique in modern reconstructive oncology. (www.actabiomedica.it)

Key words: propeller flaps, soft tissue sarcoma's defects, safety distance in resection, free flaps, reconstructive microsurgery, local tissue transfer

Introduction

Propeller flaps have been described as local flaps for soft tissue reconstruction in all areas of the body, with functional and cosmetic advantages. Historically, the nomenclature *propeller flap* was introduced for the first time in 1991 by Hyakusoku, based on a subcutaneous hub that was nourished by a central pedicle and rotated 90 degrees to correct scar contractures in extensive burn injuries. Due to its blade-shaped configuration combined with the transfer method, the flap resembles the propeller of a helicopter (1).

With the progressive understanding of the vascular system and the superior role of perforator vessels in skin flap perfusion, several and distinguished Authors reported their researches and case reports in reconstructive surgery. The main characteristic is the preservation of the major vascular axis (2-9). Another peculiarity is represented by their versatility: the flap can be selected on the perforator artery depending on the size, the location and the type of the defect (10).

The pedicle can be isolated by means of loupes, and microscope is usually not required: those flaps require a microsurgical dissection without vascular sutures; "a microsurgical not microvascular flap" as declared by Georgescu et al. (11).

Propeller flaps allow for a great freedom in design and for a wide mobilization, that extend the possibility of reconstructing defects with local tissues and minimal donor-site morbidity, avoiding the sacrifice of any unnecessary tissue, decreasing the operative time and the hospitalization. Indeed, they allow one-stage surgical procedures. The flap can be adapted to reconstructive needs, preparing a composite flap, including muscular, tendinous and skeletal components (12,13,14).

From the aesthetic point of view the final result can achieve optimal results, as the concept of "like-with-like reconstruction", minimizing the discomfort for the patient and allowing a better concealing of the scars (12-14).

Harvesting a propeller flap represents a simpler and faster alternative to free flaps; however, it requires an accurate patients' selection (history, etiology of the defects, risk factors), the preoperative planning and the dissection technique.

A hand-held Doppler probe is almost sufficient for the vascular assessment of provisional flap design: the whole region should be investigated to have a pre-operative picture of the location of all perforators and of the axial pedicle, allowing an easier identification of vessels during the dissection. (14,16) However, in cases of large flaps, CT scan is mandatory in order to perfectly study the vascular pattern of the district. In our practice, high frequency ultrasound doppler is often a hinge to decide the best perforator before surgery (Figure 1).

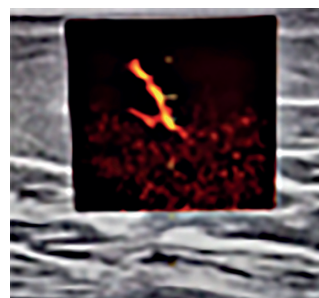


Figure 1. Preoperative planning of a propeller flap using Doppler evaluation to identify perforator vessels and mark the flap design according to the vascular pattern of the selected district.

Through optical magnification device, the supra-fascial or subfascial dissection permits a direct visualization and identification of all perforators; one is chosen on the base on its caliber, pulsatility, localization, course and orientation, number and caliber of accompanying veins (14,15). A complete dissection of the vascular pedicle from the surrounding tissue is advisable, in order to gain an adequate length and flap harvesting, avoiding any vessel buckling and torsion (16).

A crucial step remains the inseting phase, checking circulation before and after flap rotation; both clockwise and counterclockwise directions are evaluated, choosing the best one in terms of flow, rotation, color, bleeding, capillary refilling, eventually arterial spasm (14). In cases of large flaps, it is advisable a scan after the administration of green indocyanine to evaluate and localize the areas without any vascular perfusion, removing them way before the inseting (Figure 2).

The postoperative care includes several clinical observation of skin color, temperature and pulsatility; any external compression is strictly avoided. Moreover, for the first days, elevation should be promoted to help venous drainage and to control the swelling, as well as the immobilization in splints for the reconstructive surgery of the limbs (17).

Propeller flaps can be employed to gain a functional restore and an optimal cover: the surgical experience for the harvesting in loss of substance has constantly increased, including traumatic injuries, post-trauma revisions, infectious diseases, chronic ulcers, pressure sores and tumor resection (18-20,34).

The major clinical series reported in literature for upper and lower limbs have been demonstrated

positive results, as collected by Lazzeri et al. and Gir et al. (18,19).

Nevertheless, the propeller flaps present possible complications as follows: arterial insufficiency, venous insufficiency, partial necrosis, transient edema, epidermolysis, flap failure (even if rare). Application of standardized protocols and techniques can allow for a prompt identification and resolution (21). The role of specific risk factors remains undefined, such as arc of rotation, flap, size, age, sex, smoking, defect etiology, diabetes and peripheral vascular diseases (22).

The purpose of the present study was to analyze our casuistry in oncological reconstruction of a total amount of 19 patients affected by soft tissue sarcomas (STSs), and treated with propeller flaps. We compared this group of patients to an equal group treated with free microsurgical flaps during the same period of time, to enhance any differences and complications if present. Further, we analyzed the rate of complications and recurrence, with attention to the safeness and salvage rates; concluding, we briefly discuss the current literature.

Materials and methods

Thirty-eight patients affected by soft tissue sarcomas were enrolled in this retrospective study, examined between 2019 and 2023; all the patients underwent surgical reconstruction after tumor excision: nineteen microsurgical flaps and nineteen propeller flaps.

Patients between 2019 and 2021 underwent surgery at the European Institute of Oncology (IEO) and patients between 2021 and 2023 were operated at the Department of Plastic and Reconstructive Surgery, AUSL of Piacenza.

The same surgeon performed the reconstructive procedures in both groups, the same surgical demolitive technique, tips and devices were adopted in the two groups.

Concerning the site, the surgical targets were skull, upper limbs, trunk and lower limbs.

Compartmental surgery was performed according to the invasion of the tumor. Excision was performed with 3 cm macroscopic margins of healthy tissue and 1 cm as microscopic margins.

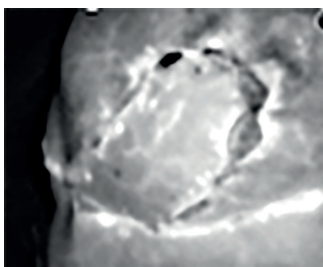


Figure 2. Intraoperative assessment of flap perfusion after rotation, using indocyanine green angiography to evaluate vascular flow and identify poorly perfused areas before final inseting.

For each harvesting, the tumor was carefully handled by sterile drapes and, after its complete excision, all the surgical gloves and instruments were changed.

After surgery, the patients were monitored at pre-established time frame by clinical observations and doppler evaluations to confirm and state the outcomes, preventing any early possible complications.

We recorded all the complications diagnosed during the hospitalization.

Finally, patients underwent radiotherapy and/or chemotherapy by a specific and custom-made oncologic program.

We reviewed similar studies in the literature, selecting meta-analysis and review and the most recent ones (23-29).

Results

21 patients were male (55.26%) and 17 were female (44.74%), with a range between 26 to 70 years old (mean age 56).

In our series 19 cases were treated harvesting free flaps (GROUP A, 50%) and 19 cases were treated with propeller flaps (GROUP B, 50%).

As free flaps we recorded as follows: 2 superficial circumflex iliac artery perforator free flaps (10.53%), 1 thoracodorsal artery perforator free flap (5.26%), 2

latissimus dorsi free flaps (10.53%), 14 anterolateral thigh free flaps (73.68%). As propeller flaps we recorded as follow: 1 internal mammary artery perforator flap (5.26%), 1 pedicle circumflex epigastric artery perforator flap (5.26%), 2 genicular flaps (10.53%), 1 thoracodorsal artery perforator flap (5.26%), 6 peroneal flaps (31.52%), 4 posterior tibial perforator-based flaps (21.05%), 2 medial sural artery perforator flaps (10.53%), 2 anterior tibial artery flaps (10.53%) [Table 1, Figures 3-8].

All the patients were hospitalized for a minimum of 4 days to a maximum of 7 days, depending on their post-operative course.

In term of complications, we recorded 1 partial necrosis (5.26%) in free superficial circumflex iliac artery perforator (SCIP) flap due to venous congestion successfully treated with leeches.

We observed 3 cases of wound dehiscence (15.79%) in propeller flaps and second intention was reached after 3 weeks. [Table 2]

Clinical and radiological follow-up was performed every 3 months for the first 2 years and every 6 months for the next 5 years. A minimum follow-up of 9 months was observed for each patient of the series. Adjuvant radiotherapy was adopted in 27 patients, according to the stage of the disease and after a multidisciplinary approach.

We faced 1 case (5.26%) of tumor recurrence in a patient treated with free flaps. [Table 2]

Table 1. Type of flaps in our series, comparing both of groups including in this study.

Type of Flaps	Group A (19, 50%)	Group B (19, 50%)
superficial circumflex iliac artery perforator (SCIP)	2 (10.53%)	-
thoracodorsal artery perforator	1 (5.26%)	1 (5.26%)
latissimus dorsi	2 (10.53%)	-
anterolateral thigh (ALT)	14 (73.68%)	-
internal mammary artery perforator	-	1 (5.26%)
pedicle circumflex epigastric artery perforator	-	1 (5.26%)
genicular flap	-	2 (10.53%)
peroneal flap	-	6 (31.52%)
posterior tibial perforator	-	4 (21.05%)
medial sural artery perforator (MSAP)	-	2 (10.53%)
anterior tibial artery flap	-	2 (10.53%)



Figure 3. Free anterolateral thigh (ALT) flap.



Figure 4. Free superficial circumflex iliac artery perforator (SCIP) flap.



Figure 5. Genicular propeller flap.



Figure 6. Thoracodorsal artery perforated free flap.

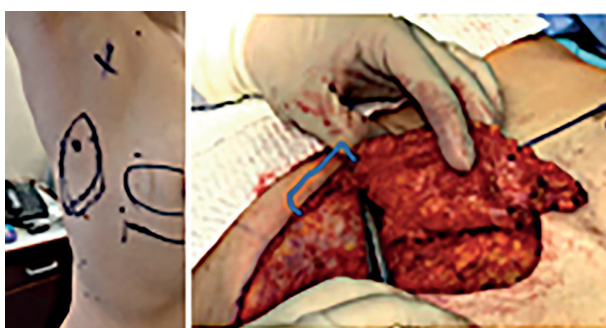


Figure 7. Thoracodorsal artery perforated propeller flap.

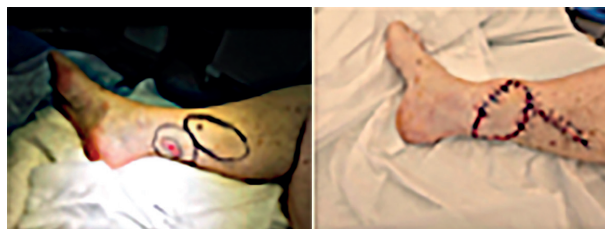


Figure 8. Posterior tibial perforated-based propeller flap.

Table 2. Index and rate of complications recorded in our series.

Complication (5, 26.32%)	Group A (19, 50%)	Group B (19, 50%)
Partial necrosis	1 (5.26%)	-
Wound dehiscence	-	3 (15.79%)
Tumor recurrence	1 (5.26%)	-
	2 (10.53%)	3 (15.79%)

Discussion

STSs are rare malignant tumors that originate from mesodermal tissues, characterized by a high local recurrence; they present most commonly as an asymptomatic solid swelling in extremities, but can occur anywhere in the body.

In past decades, the management of STSs often required extensive and disabling amputation through neoadjuvant or adjuvant therapies, improving outcomes and patients' quality of life. Current treatment has evolved toward multidisciplinary approach, including radiotherapy, local excision and plastic surgical reconstruction (23).

The surgical principles include the introduction of non-irradiated healthy tissue, a tension-free closure and the obliteration of potential dead space, ensuring both aesthetic and functional results (24). Therefore, the criterion of transferring a healthy tissue from outside the field of irradiation, has represented the gold standard until nowadays.

Radical resection and limb salvage are challenging goals for surgeons: the lack of local tissue, the bone and joint exposure prompted to consider free flaps as the first-line treatment (25).

In recent years, a greater understanding of the vascular anatomy has led to a more frequent use of perforator propeller flaps for covering the small and medium size defects, with good results comparing to free flaps (26,27).

A higher percentage of failure, partial necrosis, wound dehiscence and infection in free flap harvesting were enhanced in the meta-analysis performed by Bekara et al. (28).

Tumor spillage into the operative field is one of the most risk factors for local recurrence: as reported by Tiengo et al., the rate of recurrence at the donor site is extremely rare (only three cases described in literature), but it is mandatory to plan the side at the first biopsy inside the area that will radically resected in case of histological evidence of STSs (29).

Free margins with a cuff of healthy tissue (final margins > 1.0 cm or intact fascial plane) and minimal dissection of the surrounding tissues are estimated to be fundamental for the prevention of local recurrence (29,30).

Furthermore, in order to avoid the tumor dissemination and cells' spreading, the surgical field has to be cleaned: indeed, surgical instruments and gloves can promote a direct dissemination. After tumor excision, the chemokines and cytokines can act as wound-homing-delivery of abnormal cells (29,31,32).

Another criterion to follow in a multidisciplinary approach, is represented by the increased volume of radiotherapy focused on the proximal margin of the flap, that was nearest to the mass before surgery. In the same article, Cha et al. reported no statistical differences after surgical reconstruction between free flaps and propeller flaps, in terms of local-control and disease-free survival rate (33). Since then, no study in the literature has reported cases of recurrence involving propeller flap harvested (29).

On the base of our data, we can underline the importance of a strictly observation and practice of guidelines, precautions, and surgical refinements to eliminate any possibility of direct dissemination; notably the width of cutting margins.

As reported by Authors in other researches aforementioned, we did not register any major complications in reconstructions with propeller flaps: the

percentage rates were comparable between GROUP A and GROUP B (10.53% vs 15.79%).

However, the most severe complications occurred in patients treated with free flaps (partial necrosis and tumor recurrence).

The present paper has some limits, first of all the restricted case series; furthermore, not all the types of flaps could be observed and compared in both groups for the differences at the base of any flap pattern and vascular location. In conclusion, further investigations are advisable for an accurate analysis in the majority of cases, removing any possible statistical bias.

The surgical experience in oncological reconstruction suggests that propeller flaps achieve successful outcomes and no recurrence rate, low morbidity, quick recovery and functional/aesthetic results; we totally agree with this point of view.

Even if it is necessary to pursue a surgical training for an accurate dissection, propeller flaps are doable: this advantage can lead them to a major feasibility and wider indications. (34)

Conclusions

The data reported in literature were similar to our case series, confirming that propeller flaps represent a safe choice in oncological reconstruction if accompanied by an adequate surgical technique. Although free flaps still remain a mandatory choice for covering wide cutaneous areas and for complex defects, nonetheless, propeller ones represent a valid and feasible option for difficult wounds, and they can be raised from different body parts. These flaps provide freedom in design, low donor-site morbidity, a single-stage procedure, aesthetic results and good concealing of scars represent characteristics that should be considered for surgical treatment decisions. However, given the limited sample size, further investigations are necessary in order to provide a guidance in the choice of soft tissue reconstruction, especially after the tumor excisions.

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Authors' Contribution: Conceptualization V.N. and C.A.; Investigation V.N., S.C., E.P.; Data curation, V.N., S.C., E.P., L.R.; Writing - original draft C.A. and V.N.; Writing review V.N., C.C., F.Q., C.G.; Supervision V.N. All authors have read and agreed to the published version of the manuscript.

Ethic Approval: No approval from the Institutional Review Board or Ethics Committee was required for this study, as it is a retrospective analysis of previously treated and anonymized patients in accordance with the principles of the Declaration of Helsinki.

Conflict of Interest: Each author declares that he or she has no commercial associations (e.g. consultancies, stock ownership, equity interest, patent/licensing arrangement etc.) that might pose a conflict of interest in connection with the submitted article

Declaration on the Use of AI: Artificial intelligence tools were used only for language editing and grammar improvement.

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